

L2

CLASS X - SCIENCE

METALS AND NON-METALS

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CHEMICAL REACTIONS OF METALS



The Reactivity Series

The Reactivity Series is a list of metals arranged in the order of their decreasing reactivity.

More reactive metals displace less reactive metals from their compounds.

K	Potassium	Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
Zn	Zinc	Reactivity decreases
Fe	Iron	
Pb	Lead	
H	Hydrogen	
Cu	Copper	
Hg	Mercury	
Ag	Silver	
Au	Gold	Least reactive

Reaction with air (oxygen)

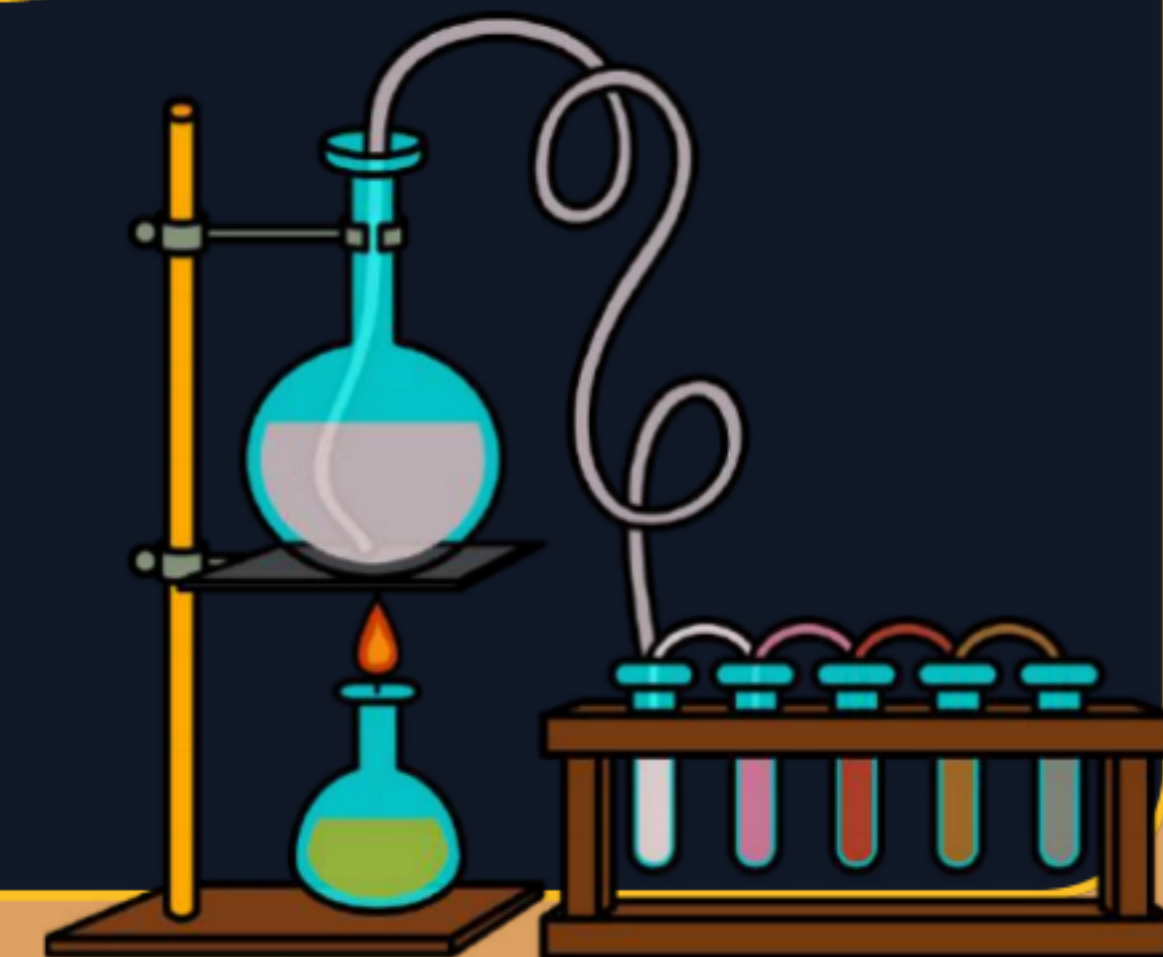
have you ever wondered why iron rusts, aluminium doesn't corrode easily, and gold always remains shiny?

When metals come in contact with oxygen present in air, they undergo oxidation and form metal oxides.



Metal oxides are generally **basic** in nature.

For example : Sodium oxide (Na_2O), Potassium oxide (K_2O), Calcium oxide (CaO)



Metal Oxide



Metal	Reaction with Oxygen	Equation	Important NCERT Points
K Na Ca Mg Potassium (K)	Reacts vigorously, catches fire if kept in open	$4K + O_2 \rightarrow 2K_2O$	Stored in <u>kerosene</u> oil to prevent accidental fires
Sodium (Na)	Very vigorous reaction	$4Na + O_2 \rightarrow 2Na_2O$	Stored in <u>kerosene</u> oil
Magnesium (Mg)	Burns with dazzling white flame → white powder	$2Mg + O_2 \rightarrow 2MgO$	<u>MgO</u> is basic oxide
Aluminium (Al)	Forms <u>Al₂O₃</u>	$4Al + 3O_2 \rightarrow 2Al_2O_3$	Q → Utensils are Made of Al? <u>Thin protective oxide layer</u> <u>prevents further oxidation;</u> <u>amphoteric</u>

Metal	Reaction with Oxygen	Equation	Important NCERT Points
<u>Zinc (Zn)</u>	Forms ZnO	$2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$	ZnO is amphoteric (yellow when hot, white when cold)
<u>Iron (Fe)</u>	Iron filings burn vigorously; forms Fe_3O_4	$3\text{Fe} + 2\text{O}_2 \rightarrow \text{Fe}_3\text{O}_4$	Iron does not burn easily as lump, but filings do
<u>Copper (Cu)</u>	Forms black coating of CuO	$2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$	Black oxide layer
<u>Silver (Ag)</u> X	No reaction	–	Does not react even at high temperature
<u>Gold (Au)</u> X	No reaction	–	Does not react even at high temperature

Amphoteric Oxides

Some metal oxides show both acidic and basic behaviour.
Such oxides, which react with both acids and bases to form salt and water, are called Amphoteric Oxides.

1. With Acid (acts as a base): $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
Base
Salt +
 2. With Base (acts as an acid): $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
Acid
Base
Salt
- Sb Zn Pb As Be Sn
- Al_2O_3 , ZnO , PbO , SnO , SnO_2 , ~~Cr_2O_3~~ , Sb_2O_3 , BeO , ~~V_2O_5~~ , ~~TiO_2~~ , ~~As_2O_3~~



Check yourself

QUES: A blacksmith can heat an iron rod for a long time without it catching fire. However, fine iron filings burn vigorously when sprinkled into a flame. Explain the reason.

Answer: Iron filings have a much larger surface area exposed to oxygen than an iron rod.

- As a result iron filings react with oxygen more rapidly and burn vigorously.
- An iron rod has compared less surface area exposed, so it reacts slowly, and it does not burn easily.



Check yourself

QUES: A student burns three metals separately in air.

Metal	Observation
A	Burns with brilliant white light.
B	slowly develops a black coating.
C	No visible change.

→ Mg

→ Cu

→ CuO

→ Ag / Au

- Arrange the metals in decreasing order of reactivity.
- Identify the black substance formed.
- Which metal is least reactive?



Check yourself

Ques: Assertion (A) : Sodium catches fire when it reacts with cold water.

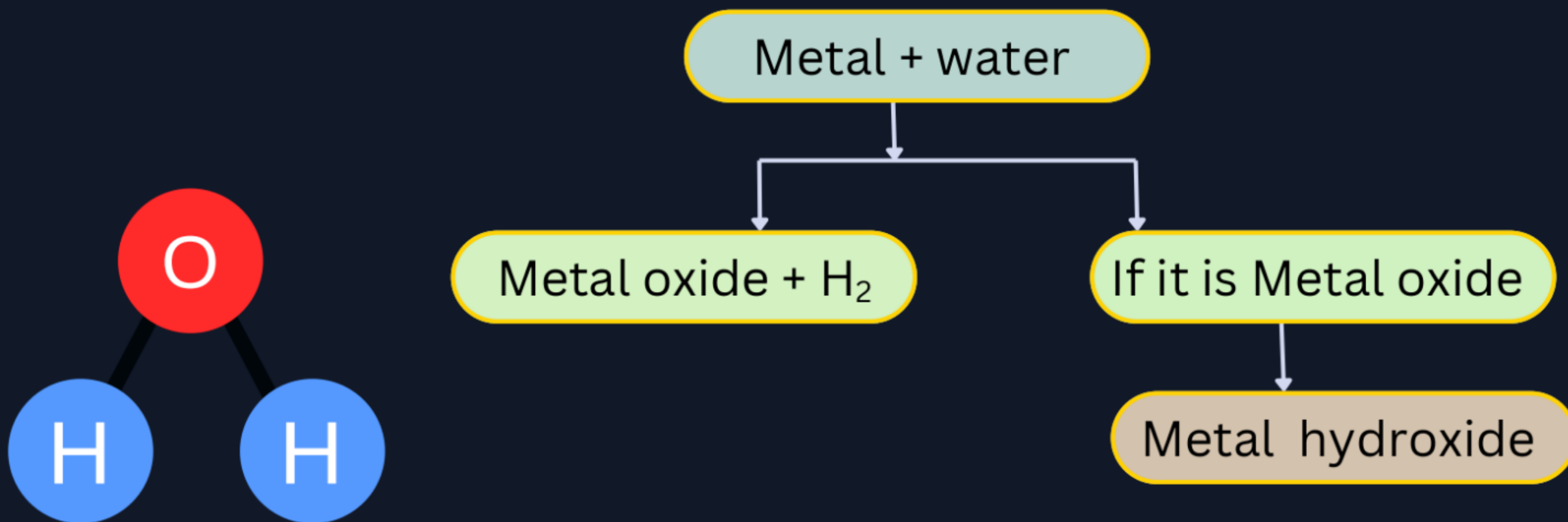
Reason (R): The reaction between sodium and cold water is highly exothermic.

- ☒ a) Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, and R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.



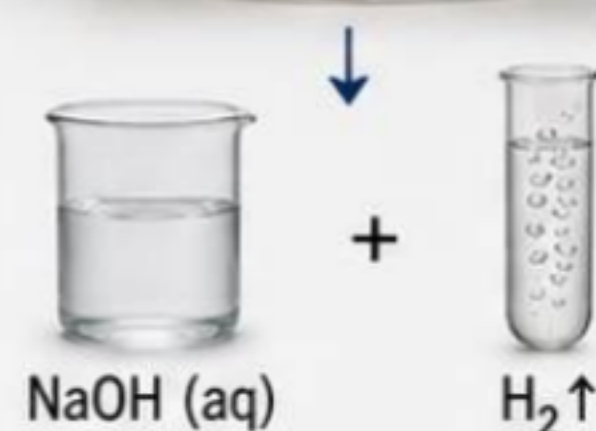
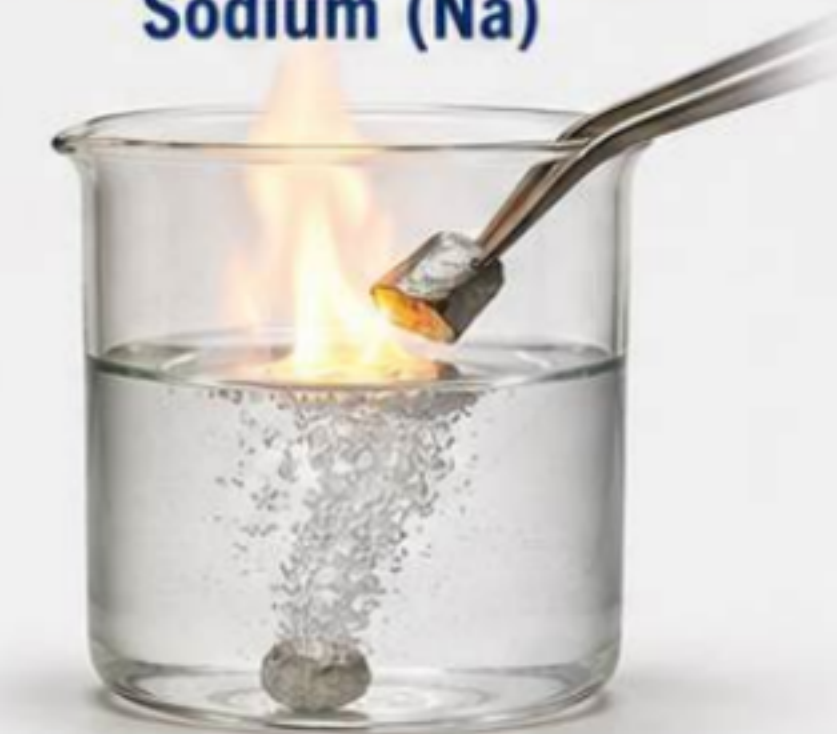
Reaction of Metals with water

Metals react with water to form metal oxides (metal hydroxide) and hydrogen gas. However, all metals do not react with water.

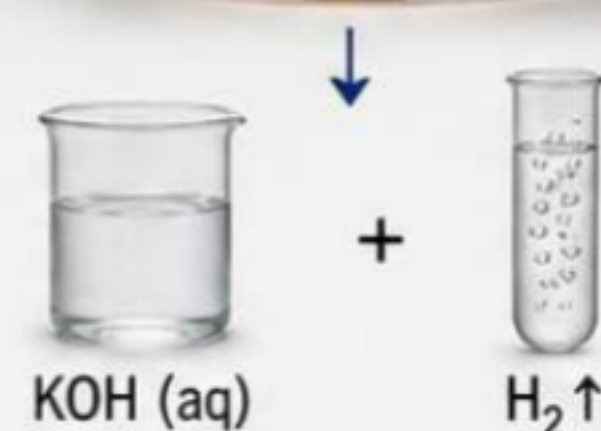
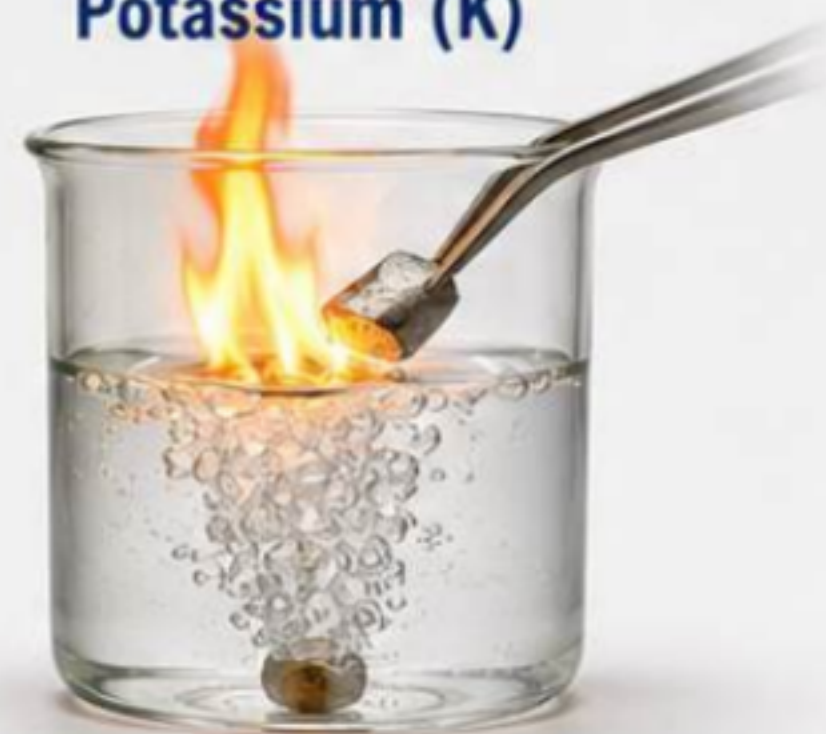


REACTION OF METALS WITH COLD WATER

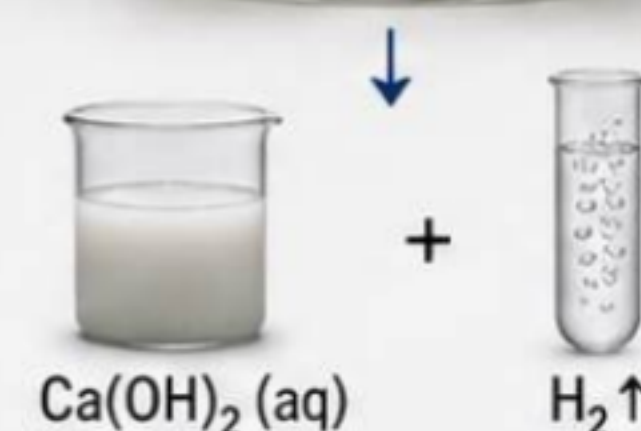
Sodium (Na)



Potassium (K)

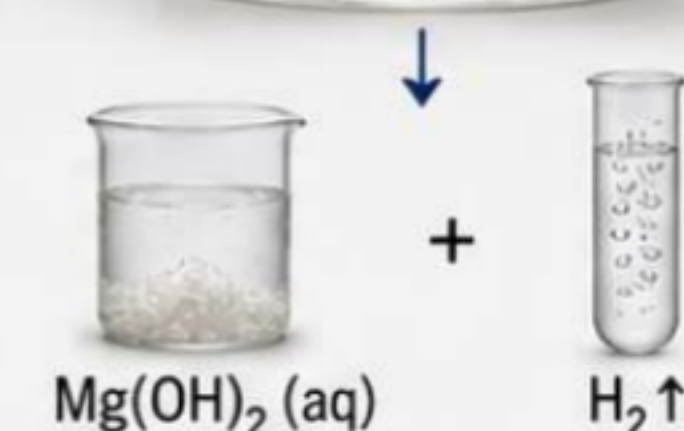


Calcium (Ca)



Magnesium (Mg)

With hot water



NO REACTION WITH WATER (COLD OR HOT)

Lead (Pb)



×

Copper (Cu)



×

Silver (Ag)



×

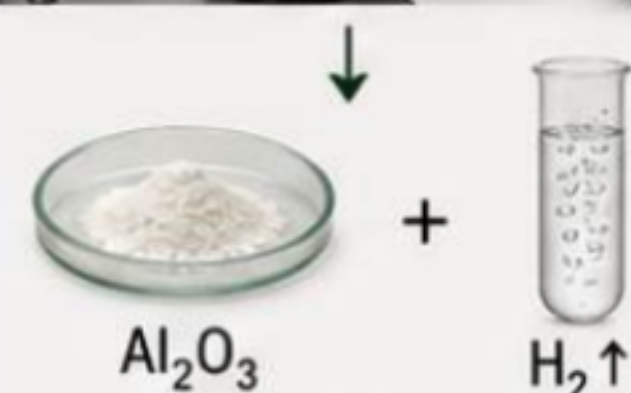
Gold (Au)



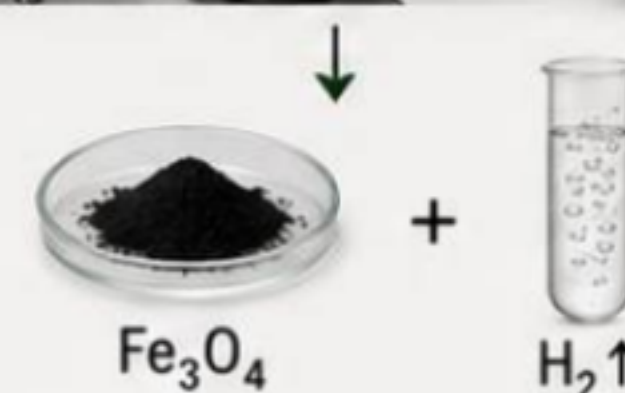
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REACTION WITH STEAM (HOT WATER VAPOUR)

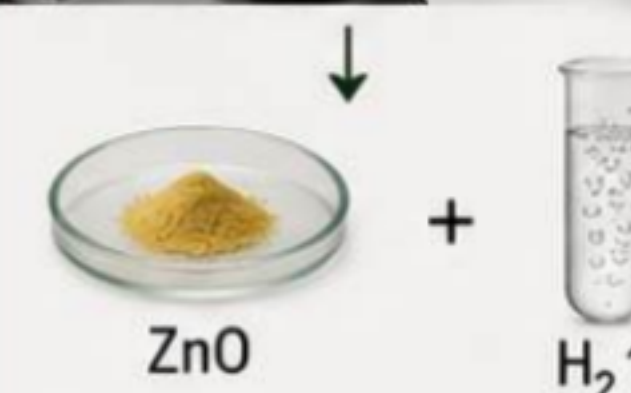
Aluminium (Al)



Iron (Fe)



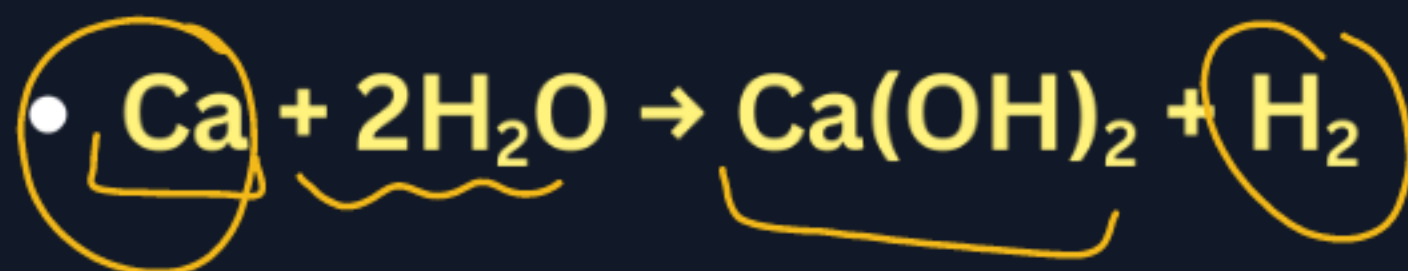
Zinc (Zn)



Alkali Metals (Na, K, etc.): Alkali metals are very reactive metals (like sodium and potassium) which react vigorously with oxygen and water to form strong soluble bases (alkalis). They are stored in kerosene to prevent accidental fires.

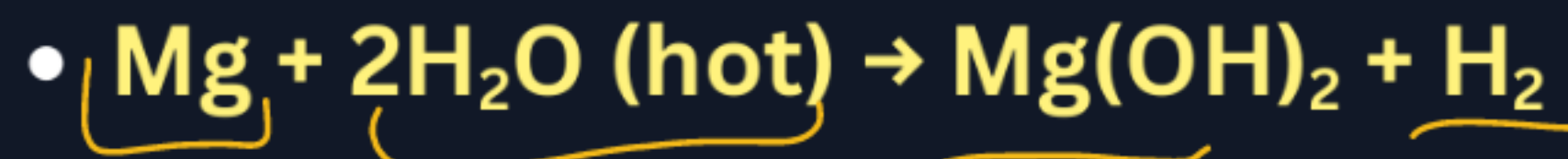


(Reaction is so violent and exothermic that hydrogen catches fire).



- Reacts less violently with cold water.
- Hydrogen does not catch fire.
- Calcium floats (due to H_2 bubbles sticking to surface).

With hot water:



Mg
Also floats (H_2 bubbles stick to surface)
No reaction with cold water.

React with steam:

- $2\text{Al} + 3\text{H}_2\text{O (steam)} \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$
- $3\text{Fe} + 4\text{H}_2\text{O (steam)} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
- $\text{Zn} + \text{H}_2\text{O (steam)} \rightarrow \text{ZnO} + \text{H}_2$

No reaction with cold/hot water.

No Reaction

Do not react with water at all.



K }
Na } → Cold

Ca → Cold

Mg → Hot

Al }
Zn } → Steam
Fe }

Pb }
H }
Cl }
Ar } → NO
Ag }
V }

DID YOU KNOW?

Science spotlight

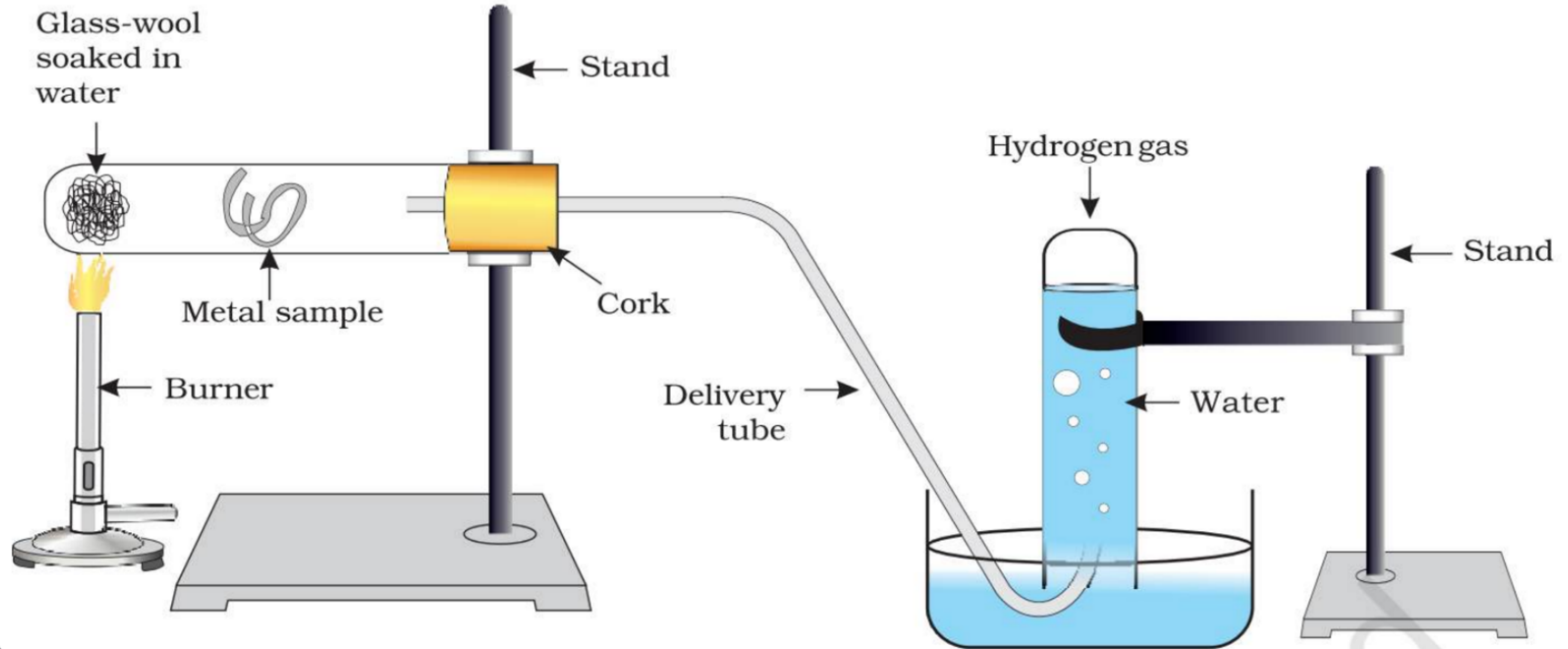
Why do some metals float in water??

Calcium and magnesium start floating on water because hydrogen gas bubbles formed during the reaction stick to their surface, making them bouyant.



Floating doesn't mean the metal is lighter than water.

It floats because of hydrogen gas bubbles reduce its effective weight in water.



Check yourself

QUES. A student performs an experiment by adding four different metals separately to water. The observations are recorded below.

Metal Observation → Na, K

A. Reacts violently with cold water and catches fire. → Ca

B. Reacts slowly with cold water and starts floating on the surface.

C. Does not react with cold water but reacts with steam to produce hydrogen gas.

D. Does not react with water under ordinary conditions. → Al, Zn, Fe

The metals are sodium (Na), calcium (Ca), iron (Fe), and copper (Cu). → Cu, Ag, Au

Answer the following questions:

- (a) Identify metals A, B, C, and D.
- (b) Write the balanced chemical equation for the reaction of
 - (i) Metal A with water
 - (ii) Metal C with steam
- (c) Name the gas evolved during these reactions.
- (d) Arrange the four metals in decreasing order of their reactivity.
- (e) State one reason why Metal A is stored under kerosene.



Check yourself

QUES: A student performs three experiments by adding water separately to

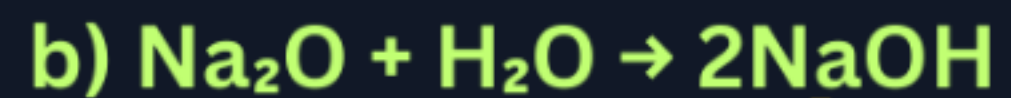
- Sodium oxide (Na_2O)
- Magnesium oxide (MgO)
- Copper (II) oxide (CuO)



Answer the following:

- Which oxide(s) form an alkali on reacting with water?
- Write the balanced chemical equation?
- Why does copper(II) oxide not produce an alkaline solution?

Answer: a) Na_2O and K_2O form alkalis.



c) Copper(II) oxide is insoluble in water, so it does not react with water to produce an alkaline solution.



Check yourself

→ K, Na

Q. Metal X reacts vigorously with cold water, producing a gas that catches fire.

Metal Y reacts less violently with cold water and starts floating. → Ca

Metal Z does not react with cold water but reacts with hot water to form a sparingly soluble hydroxide.

Identify X, Y and Z:

☒ (a) X = Sodium, Y = Calcium, Z = Magnesium

(b) X = Potassium, Y = Sodium, Z = Iron

(c) X = Sodium, Y = Magnesium, Z = Calcium

(d) X = Calcium, Y = Sodium, Z = Zinc

→ Mg

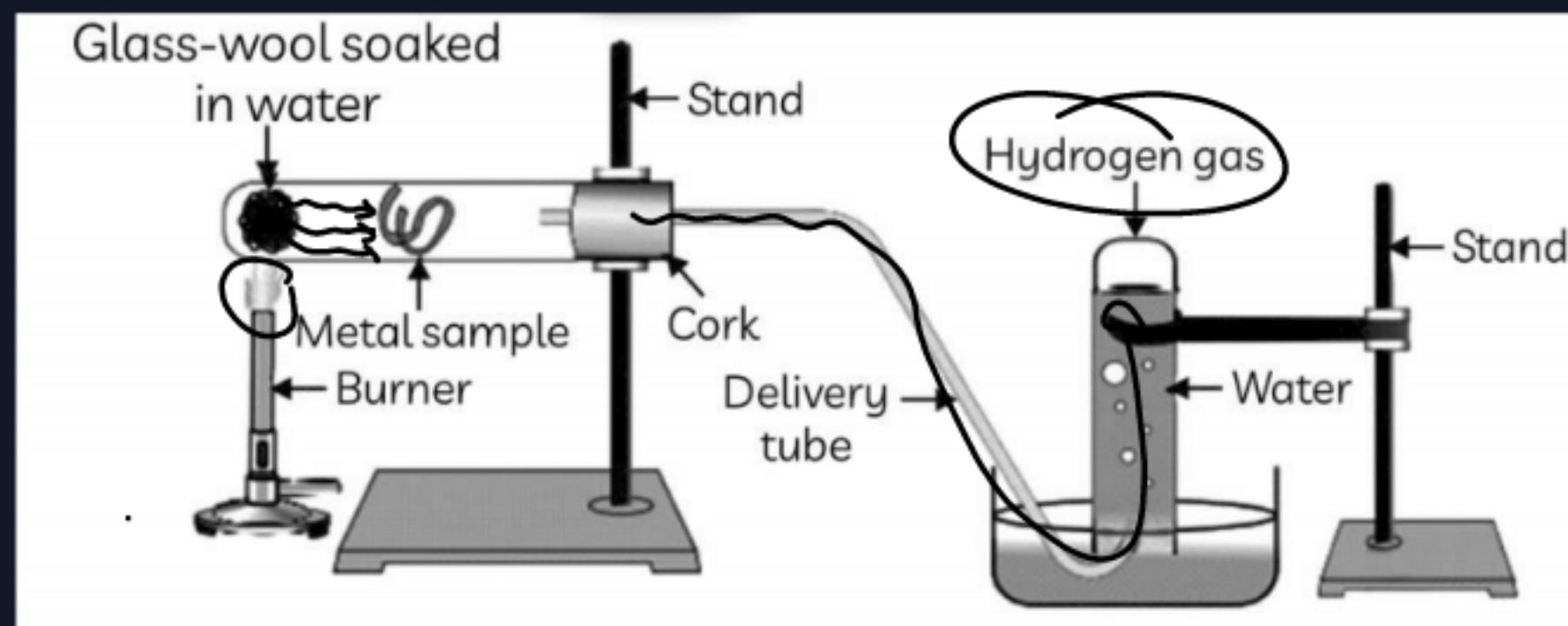
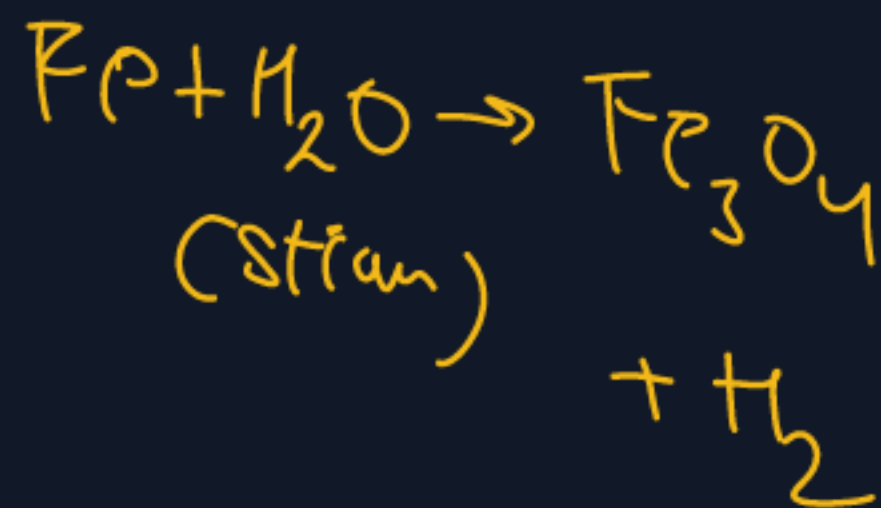


Check yourself

Q. A teacher performed an experiment where he collected the following sample of metals: Aluminium, copper, iron, lead, magnesium, zinc and sodium. He put the small pieces of these samples into separate beakers half-filled with cold water and noticed the reactions. Later, those metals who did not react with cold water at all were transferred to another beaker half-filled with hot water. After that, the teacher set up the following apparatus for the metals that did not react even with hot water.

(A) From the given samples of metals, identify the ones which did not react with either hot or cold water.

(B) From the given figure, what could you infer about the chemical properties of metals identified in (A). Support your answer by providing any one chemical equation regarding the same.





Reaction of metals with Acids

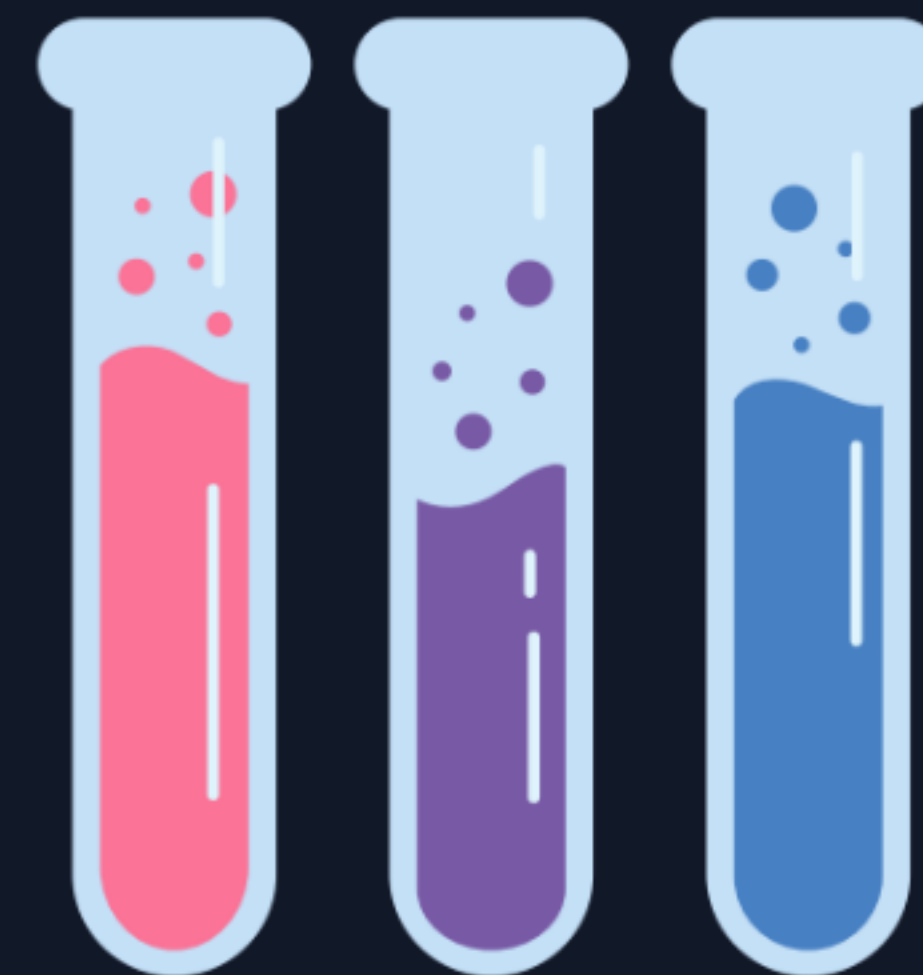
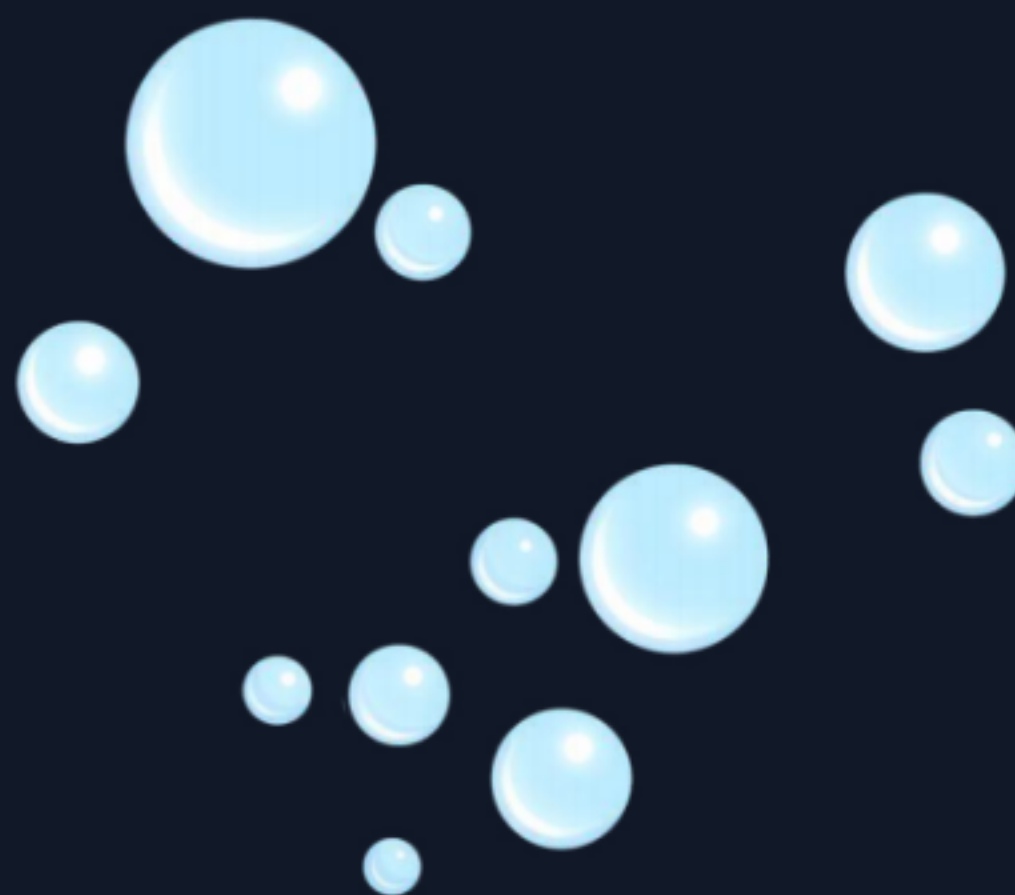
Most metals react with dilute acids to form a **salt and hydrogen gas**.

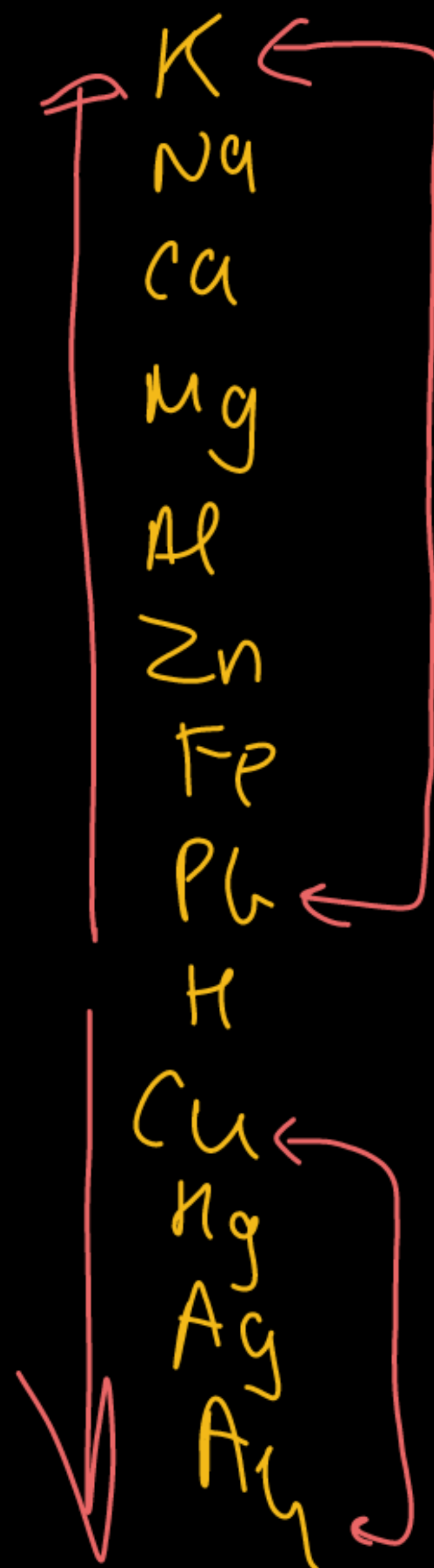
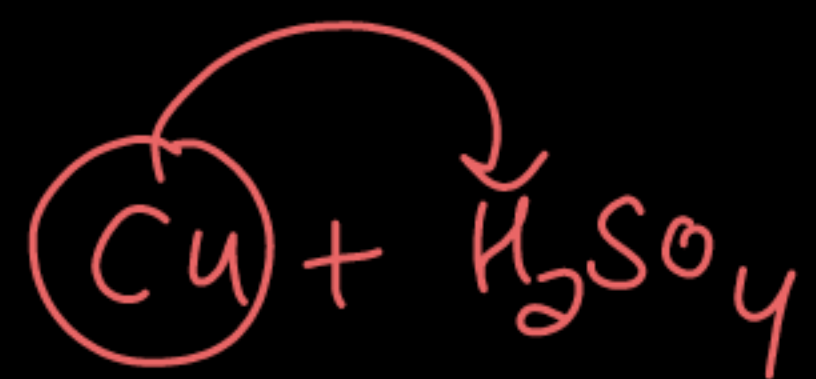


Metal + dilute acids $\rightarrow$ Salt + H_2

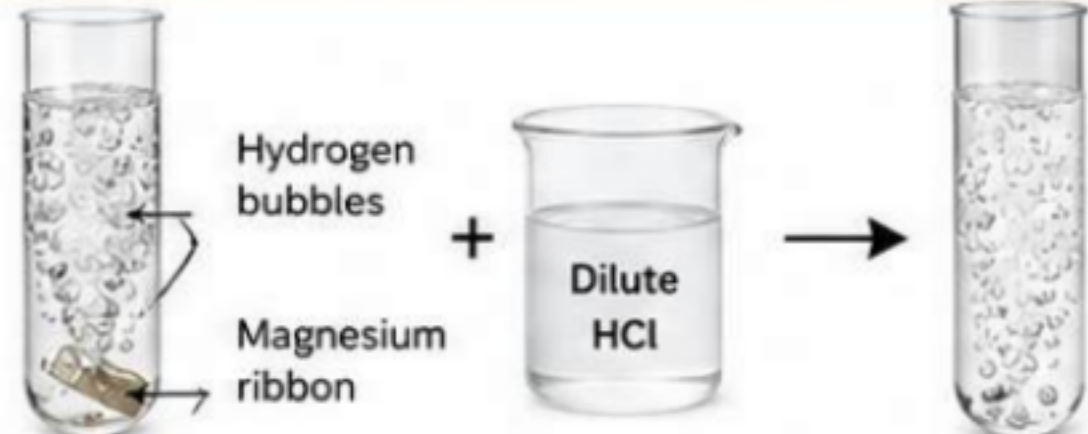
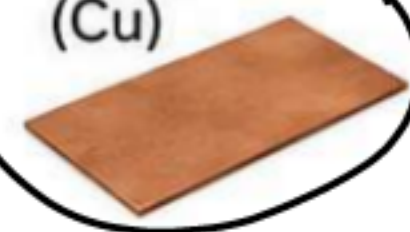
Observation

- ✓ Hydrogen gas is evolved.
- ✓ The gas burns with a pop sound when tested with a burning splint.







METAL	REACTION WITH DILUTE HCl	CHEMICAL EQUATION	OBSERVATIONS	CONCLUSION
1. Magnesium (Mg) 		$\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$	- Reacts most vigorously. - Fastest bubble formation. - Highest temperature rise.  Highest temperature rise	Magnesium is most reactive with dilute HCl.
2. Aluminium (Al) 		$2\text{Al} + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2$	- Reacts vigorously, but less than Mg. - Good bubble formation.  High temperature rise	Aluminium is very reactive but less than magnesium.
3. Zinc (Zn) 		$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$	- Moderate reaction. - Moderate bubble formation.  Moderate temperature rise	Zinc reacts with dilute HCl but less vigorously.
4. Iron (Fe) 		$\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$	- Slower reaction. - Less bubble formation. - Small temperature rise.  Small temperature rise	Iron reacts slowly with dilute HCl.
5. Copper (Cu) 		X No Reaction	- No bubbles observed. - No temperature change.  No temperature change	Copper does not react with dilute HCl.

Reactivity Order with Dilute HCl: $\text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > \text{Cu}$ (No reaction)

Bye Bye :)

